

Khalil Belharir

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Profile

Software engineering candidate with strong foundations in computer science, systems programming, cloud infrastructure, testing, and Python/Java/C++/C# development. Previous engineering experience at Amazon in Dublin and Société Générale, focused on building reliable tools, automating workflows, and solving ambiguous technical problems.

Education

Ensimag, Grenoble Institute of Technology

Dual Master's Degree in Computer Science, Engineering & Quantitative Finance

Grenoble, France

2023 – Apr 2026

Relevant coursework: Software Engineering, OOP, Advanced, Algorithms, Operating Systems, Concurrent & Parallel Computing, Distributed Systems, Databases, Computer Architecture, Networks, Security, Programming Language Theory, Machine Learning, NLP, Statistics, Time Series.

Panthéon-Sorbonne University

BSc in Applied Mathematics, Economics & Computer Science

Paris, France

2019 – 2023

First-Class Honours, GPA 4.0, ranked **2/158**. Coursework: Optimization, Measure Theory, Differential Calculus, Statistics, Stochastic Calculus, Computer Science.

Lycée La Bruyère

Preparatory Classes

Versailles, France

2019 – 2022

Highly selective intensive mathematics program preparing for competitive national entrance exams. Coursework: Mathematics, Computer Science, Probability, Statistics.

Experience

Société Générale

Machine Learning / Software Engineering Intern

Paris, France

Apr – Oct 2026

- Designed a machine learning service to forecast compute infrastructure load and support real-time scaling decisions.
- Developed an uncertainty-aware predictive autoscaling approach inspired by MagicScaler, combining workload time-series forecasting, multi-scale feature extraction, and probabilistic risk estimation.
- Formulated scaling decisions as an optimization problem balancing idle resource cost, worker launch latency, and capacity/QoS risk under stochastic demand.
- Built a separate C#/.NET orchestration platform to automate the onboarding and execution of Quantitative Investment Strategies (QIS) from declarative YAML configurations.
- **Tech:** Python, PyTorch, Time Series Forecasting, C#, .NET, Temporal, REST APIs, Kubernetes, Azure resources.

Amazon

System Development Engineer Intern

Dublin, Ireland

Jun – Sep 2025

- Built cloud infrastructure tooling for non-AWS Amazon services including Retail, Kindle, Twitch, and internal platforms.
- Designed an automated test-resource generation and cleanup tool, reducing provisioning/deletion time by **96–98%**.
- Developed parallelized resource-management logic in object-oriented Python with unit tests and API integrations.
- **Tech:** Python, Unit Tests, Parallel Computing, OOP, APIs, AWS resources.

Selected Projects

Systems Programming: Implemented low-level systems projects including a RISC-V CPU design in VHDL, RISC-V assembly programs, a C interpreter, and a JPEG encoder. **Tech:** C, VHDL, RISC-V assembly.

Compiler Construction: Built a compiler for Deca, a Java-like language targeting IMA virtual-machine assembly, including parsing, semantic analysis, and code generation. **Tech:** Java.

Database Systems: Designed and implemented a relational database for an auction platform, including schema design and application-level interactions. **Tech:** SQL, Java, Oracle.

Optimization & Simulation: Developed simulation and optimization projects including firefighter robot scheduling and signal/image processing tools for satellite images. **Tech:** Java, Python, NumPy.

ML & Time Series: Implemented ARMA, GARCH, and MIDAS models for forecasting and volatility modeling, including MLE estimation and simulation tools. **Tech:** R, C++.

Skills

Programming: Python, Java, C/C++, C#, SQL, R, Unix/Linux

Software Engineering: Unit Testing, Object-Oriented Programming, APIs, Git, Software Design, Code Quality

ML / Data: PyTorch, Scikit-learn, NumPy, Pandas, Time Series, NLP, Neural Networks, Statistical Modeling

Systems: Algorithms, Data Structures, Parallel Computing, Distributed Systems, Operating Systems, CPU Architecture

Mathematics: Probability, Statistics, Optimization, Numerical Methods, PDEs, Information Theory

Languages: French: Native English: C1 Advanced, TOEIC 955 Spanish: B1 Mandarin: A2 Arabic: Basic